

CHUMPHON, Thailand

WMO: 485170

Lat: 10.4990N

Lon: 99.1884E

Elev: 6

StdP: 101.25

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) 19.9	(c) 21.0	(d) 16.5	(e) 11.8	(f) 22.9	(g) 17.9	(h) 12.8	(i) 23.8	(j) 8.1	(k) 26.9	(l) 7.4	(m) 27.3	(n) 0.1	(o) 300	(p) 0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 4	(b) 8.7	(c) 34.9	(d) 26.7	(e) 34.2	(f) 26.6	(g) 33.5	(h) 26.4	(i) 28.0	(j) 32.4	(k) 27.6	(l) 32.0	(m) 27.2	(n) 31.7	(o) 2.6	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.0	22.7	30.3	26.2	21.7	30.0	26.0	21.4	29.9	90.0	32.3	87.7	32.4	86.2	31.9	30.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min		Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
6.7	5.7	5.0	DB	17.6	36.5	1.8	1.1	16.3	37.3	15.2	37.9	14.2	38.5	13.0	39.3	
			WB	16.3	29.0	1.6	0.7	15.2	29.5	14.2	29.9	13.3	30.3	12.1	30.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	26.2	27.1	28.3	29.3	29.0	28.4	27.9	28.0	27.8	27.5	26.9	26.0
	DBStd	1.55	1.24	1.12	1.15	1.17	1.31	1.23	1.22	1.18	1.06	1.12	1.18	1.42
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6457	502	478	566	579	588	553	556	557	533	541	506	497
	CDD18.3	3416	244	244	308	329	330	303	297	299	283	283	256	239
	CDH23.3	34465	2152	2474	3443	3931	3659	3222	2962	2987	2708	2618	2302	2007
Wind	WSAvg	1.4	1.4	1.3	1.2	1.1	1.4	1.6	1.6	1.7	1.5	1.2	1.3	1.6
	PrecAvg	2331	65	40	108	99	234	253	285	297	272	294	248	136
	PrecMax	2898	311	204	510	230	451	447	481	752	442	614	534	494
	PrecMin	1741	4	0	1	1	104	84	104	120	140	86	55	0
	PrecStd	299	61	49	137	69	90	83	97	131	77	116	126	127
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.5	34.0	35.3	36.2	35.7	34.7	34.0	33.7	33.6	33.2	32.9	32.4
		MCWB	25.0	25.5	25.7	26.8	27.0	26.7	26.1	25.9	26.4	26.3	26.1	25.4
	2%	DB	31.5	32.7	34.1	35.1	34.5	33.7	33.0	32.9	32.7	32.3	31.9	31.3
		MCWB	25.3	25.6	26.2	26.8	27.0	26.4	25.9	25.9	26.1	26.3	26.0	25.4
	5%	DB	30.8	32.0	33.3	34.3	33.6	32.8	32.2	32.1	31.9	31.6	31.1	30.5
		MCWB	25.1	25.4	26.1	26.8	26.9	26.2	25.9	25.9	26.0	26.2	26.0	25.2
	10%	DB	30.1	31.1	32.4	33.5	32.6	31.9	31.2	31.2	30.9	30.8	30.2	29.8
		MCWB	25.0	25.3	26.0	26.8	26.7	26.1	25.9	25.8	25.8	26.0	25.7	24.8
	0.4%	WB	26.7	27.1	27.7	28.5	28.7	28.2	27.7	27.6	27.6	27.6	27.5	27.1
		MCDB	29.9	31.6	32.3	33.8	32.9	31.8	30.9	31.2	31.1	30.7	30.1	30.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	26.2	26.5	27.1	27.8	28.0	27.3	26.9	26.8	26.8	27.0	26.9	26.5
		MCDB	29.9	31.0	32.1	33.2	32.7	31.5	31.0	30.9	30.9	30.9	30.3	29.8
	5%	WB	25.8	26.1	26.7	27.5	27.4	26.7	26.4	26.3	26.4	26.6	26.5	26.0
		MCDB	29.5	30.6	31.6	32.7	32.1	31.0	30.4	30.4	30.3	30.4	29.9	29.3
	10%	WB	25.4	25.7	26.4	27.1	26.9	26.3	26.0	26.0	26.0	26.2	26.0	25.5
		MCDB	29.0	30.0	31.2	32.3	31.5	30.4	30.0	30.0	29.8	29.8	29.4	28.6
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	8.8	8.7	7.5	6.9	6.6	6.5	6.6	6.5	6.6	7.5
		MCDBR	9.0	9.7	10.1	9.9	8.9	8.4	8.1	7.9	7.9	8.0	8.1	8.5
		MCWBR	3.8	3.7	3.7	3.4	3.0	2.7	2.6	2.6	2.7	3.1	3.3	3.6
	5% WB	MCDBR	8.0	8.7	9.1	9.2	8.3	7.6	7.2	7.1	7.2	7.3	7.2	7.4
		MCWBR	3.7	3.7	3.7	3.6	3.3	3.0	2.9	2.8	2.9	3.2	3.3	3.5
Clear-Sky Solar Irradiance	taub		0.495	0.515	0.520	0.483	0.454	0.451	0.450	0.444	0.430	0.462	0.484	0.483
	taud		2.152	2.096	2.124	2.243	2.370	2.410	2.396	2.405	2.437	2.322	2.230	2.206
	Ebn at Noon		814	813	814	835	844	836	840	857	879	848	818	815
	Edh at Noon		152	166	163	144	124	117	120	121	118	130	139	141
All-Sky Solar Radiation	RadAvg		4.81	5.48	5.74	5.82	4.99	4.65	4.42	4.56	4.65	4.35	4.27	4.37
	RadStd		0.37	0.32	0.44	0.40	0.44	0.25	0.40	0.45	0.35	0.42	0.43	0.37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.23	N/A	N/A	N/A	+0.27	+0.31	N/A	N/A	N/A	N/A
Regional (6 neighbors)	+0.19	+0.43	N/A	+0.37	+0.30	+0.30	N/A	N/A	+76	+76

Nomenclature: See separate page